



Palliative Care - Continuous Subcutaneous Infusion Procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All services	Nursing, Medical, Pharmacy

2. Purpose

The subcutaneous route is effective and minimally invasive which is especially suitable in patients with swallowing and malabsorption issues. The subcutaneous pump (Niki T34™ pump) is small, portable, and minimally intrusive promoting optimal patient-centred care.

The administration of medication in palliative care via the subcutaneous route improves patient symptom management and comfort through the provision of medications at a constant rate. Subcutaneous administration provides more stable levels of medication and avoids the peaks and troughs of intermittent administration.

3. Equipment (remove if not relevant)

- Saf-T-Intima™ catheter
- Niki T34™ pump
- Niki T34™ lockable box (Large Niki T34™ lockable box if using 50ml syringe)
- Luer lock extension tubing
- Tegaderm (6cm x7cm) or other occlusive film dressing
- 20ml BD Plastipak syringe (30mL BD Plastipak syringe or 50ml Terumo syringe if specially ordered)
- Needle free IV cap
- Chlorhexidine 1% in isopropyl alcohol 70% swab
- Subcutaneous additive label and extension tubing label



Figure 1: Niki T 34™

Figure 2: Subcutaneous Infusion Labels

For Subcutaneous Use Only

Patient ID: _____ DOB: _____

Medication/s	Amount (units)	÷	Volume (mL)	=	Conc (units/mL)

Diluent: _____

Date: _____ Time: _____

Prepared by: _____ Checked by: _____

Subcutaneous

Subcutaneous Commenced: Date: ____/____/____ Time: ____:____:____

4. Procedure

4.1 Setting up a new pump

Medical staff

1. Prescribe appropriate medications and diluent for the pump on the AKMR125 chart.
2. No more than three (3) medications should be used in any one syringe driver.
3. Ensure the compatibility of multiple medications and diluent is checked prior to prescribing.
4. Pumps are automatically set to deliver medication over a 24-hour period.

Ensure that the patient and/or family is educated on the reason for the Continuous Subcutaneous Infusion pump and how it operates

Nursing staff

1. Gather all equipment and perform hand hygiene.
2. Accurately draw up all medications.
3. Then add diluent to the syringe and gently mix
 - 3.1. If using a 20ml syringe add diluent to a total volume of exactly 18ml.
 - 3.2. If using a 30ml syringe add diluent to a total volume of exactly 23ml.
 - 3.3. If using a 50ml syringe add diluent to a total volume of exactly 32ml.

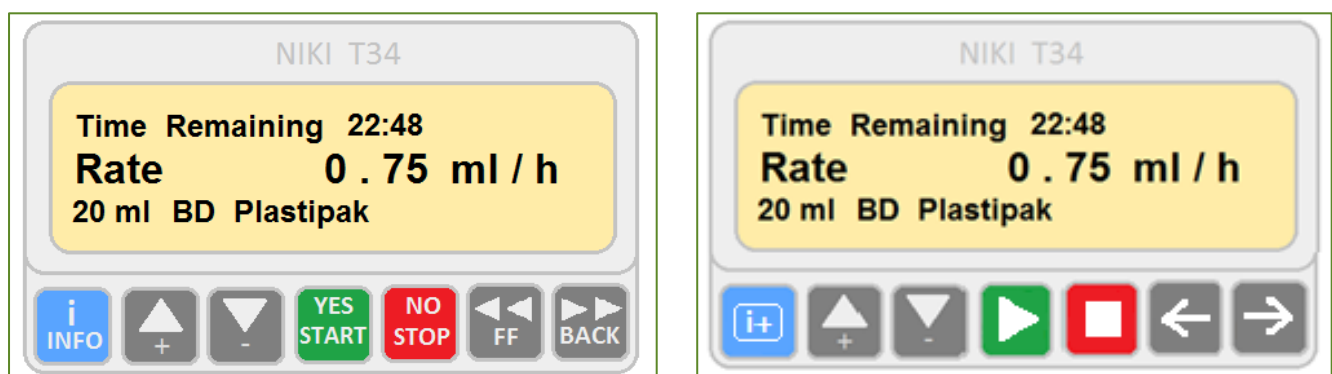


Figure 3: Niki T34™ Setup Controls (older and newer models)

4. Battery test for the Niki T34™ pump:

- Switch the pump on. To turn NIKI T34™ on, press and hold the ON/OFF button
 - Press blue INFO key
 - Select BATTERY LEVEL from the menu and press green/YES key to confirm
 - Verify battery charge is above 25% and/or sufficient to complete current program If not, change the battery
 - Turn pump off. Press and hold the ON/OFF button
5. Label the syringe as per [AKG Medication Management Procedure](#).
6. Insert the syringe into the Niki T34™ pump.

The extension line and Saf-T-Intima™ catheter must be attached and primed through the pump, with the medication in the syringe.

Do not prime manually or with any other solution

7. To start pump, press and hold down ON/OFF button on the NIKI T34™ pump.
- Lift and turn barrel clamp handle, insert prepared syringe into the correct position. If the actuator is not in the correct position to accommodate the syringe, use the FF/→ or BACK/← buttons to adjust. Close barrel clamp handle.
 - Check screen. Confirm or adjust syringe type, Press green/YES.
 - Check volume, duration and rate. Confirm, press green/YES if correct.
 - Display will read “Start Infusion?” press red/NO.
 - Press FF to PURGE (prime line).
 - To purge, press green/YES then press and hold down FF key until prime complete. Note purge volume on AKMR125 chart as volume infused.
8. To insert the Saf-T-Intima™ catheter:



Figure 4: BD Saf-T-Intima™

8.1. Perform hand hygiene.

- Select and cleanse appropriate insertion site (see Figure 5), allow to air dry.
- Unsuitable areas include:
 - Lymphoedematous areas (as impaired absorption and potential introduction of infection)
 - Sites over bony prominences/joints (may be insufficient subcutaneous tissue)
 - Previously irradiated sites (as impaired absorption)
 - Sites near a joint (movement may cause cannula to become dislodged)
 - Areas of infection/inflammation (risk of malabsorption, tissue trauma and infection) and
 - Areas with broken skin
- Remove and dispose of white tubing clamp and plastic needle cap.

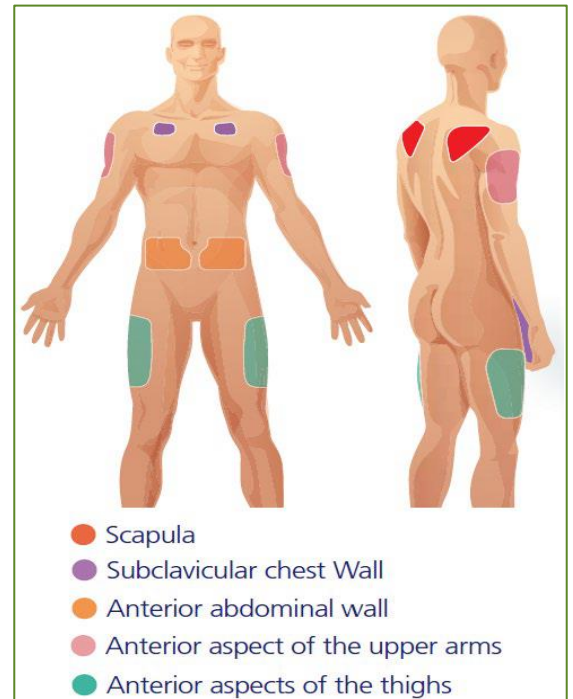


Figure 5: Subcutaneous Placement Sites

- Grip the bubbled surface of the yellow wings of the Saf-T-Intima™ catheter between the thumb and the forefinger.
- Ensure the bevelled edge is orientated upwards with the point towards the skin.
- Hold the skin between the thumb and forefinger and insert needle at a 30-45° angle.
- Secure the wings against the skin and remove the introducer ensuring that the needle is covered after removal.
- Cover with an occlusive film dressing. Document the date and time on the film dressing label. Ensure the insertion site is visible as regular site checks are required.

9. To start the Niki T34™ pump:

- Reconfirm syringe type by pressing green/YES.
- Display will read “YES to Resume, NO for New Syringe”. Press green/YES.
- Check volume, duration, and rate. Confirm, press green/YES.
- Display will read “Start Infusion” press green/YES.
- Pump will display “Pump Delivering”.

10. Lock pump by pressing and holding blue information button. Ensure Perspex box is locked.

- Document the insertion site of the Saf-T-Intima™ catheter in the patients progress notes and care plan.
- Ensure that pump details are documented on the AKMR125 form

4.2 Changing the syringe

1. If any change to medication contents or dosage is prescribed the infusion must be set up as a new pump and extension line.

2. The existing Saf-T-Intima™ catheter may be used provided the insertion site is free from signs of infection and inflammation.

The Saf-T-Intima™ catheter must be re-sited every seven days or earlier if signs of infection, irritation or swelling are present at the site of the catheter.

If the Saf-T-Intima™ catheter needs to be changed; the line and syringe must also be replaced.

3. If no change to medication or dosage is made, use the existing extension line providing it has not been in use for more than 72 hours.

The extension line must be changed at a minimum every 72 hours.

4. Prepare syringe as per section 4.1.
5. Press red/STOP button. Turn pump off using ON/OFF button to reset. Open locked box.
 - Remove old syringe and attach new syringe to extension line using non-touch technique.
 - Turn pump on and load syringe as described in section 4.1.
 - Confirm syringe type and press green/YES.
 - Display will read “YES to Resume, NO for New Syringe”. Press red/NO.
 - Check volume, duration and rate. To confirm, press green/YES.
 - Display will read “Start Infusion?” press green/YES.
 - Pump will display “Pump Delivering”.
6. Lock pump by pressing and holding blue information button. Ensure Perspex box is locked
7. Document in the patient’s medical record and on the AKMR125 form.

4.3 Monitoring the infusion

1. Monitoring of vital signs as indicated by patient condition.
2. Record volume infused (VI), volume to be infused (VTBI), infusion rate, battery level and time remaining, hourly for the first two (2) hours for a new infusion or if new medication is added.
3. Continue recording volume infused (VI), volume to be infused (VTBI), infusion rate, battery status and time remaining, four (4) hourly.
4. Record pain, nausea, symptom scores at rest and on movement. Omit if asleep.
5. Pain, 0 = no pain 10 = worst pain; Nausea, 0 = no nausea 1= nausea present.
6. If there are other symptoms for example secretions, note them in the comments box.
7. Observe the Saf-T-Intima™ catheter site for signs of infection or leaking each time monitoring is recorded. If present, the catheter must be removed and a new one re-sited. The catheter may remain insitu for 7 days if no signs of infections are present.

5. Compliance monitoring

Compliance against this policy will be evaluated with annual auditing of the Palliative care unit subcutaneous infusion chart.

6. Policy context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [High Risk Medication policy MP 0131/20](#)
- HDWA [Consent to Treatment Policy OD 0657/16](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching EMHS: 21](#)

7. Supporting information

The following documents support the implementation of this procedure:

- EMHS [Consent Policy EMHS: 5](#)
- [Medication Policy AKG-POL-0253-13](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)
- [Pain Management Procedure AKG-PRO-0339-17](#)
- Armadale Kalamunda Group AKMR 125, Palliative Care Unit subcutaneous infusion form

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.5 Healthcare rights and informed consent.
- Standard 3 – Action 3.8 The health service organisation has a hand hygiene program.
- Standard 3 - Action3.9 The health service organisation has processes for aseptic technique.
- Standard 3 - Action3.10 The health service organisation has processes for the appropriate use and management of invasive medical devices that are consistent with the current edition of the [Australian Guidelines for the Prevention and Control of Infection in Healthcare](#)
- Standard 4 – Action 4.3 Partnering with consumers
- Standard 4 – Action 4.15 High-risk medications
- Standard 5 – Action 5.16 Comprehensive care at end of life
- Standard 5 – Action 5.20 Clinicians support patients, carers and families to make shared decisions about end-of-life care in accordance with the *National Consensus Statement: Essential elements for safe and high-quality end-of-life care*²²⁰

9. Document contact

Enquiries relating to this procedure may be directed to:

Title: Nurse Unit Manager
Department: Palliative Care Unit, Kalamunda Hospital

10. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
v2	09/02/2024	09/02/2027	Minor amendment

11. Authorisation

This procedure has been authorised and issued by:

Authorised by	Tania Strickland – A/Director Clinical Services
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